

LOCHAN RAJ DAHAL

Agentic Systems Builder · Full-Stack Developer · Technical Educator
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PROFILE

Self-taught builder and technical educator based in Nepal with 5 years of classroom experience teaching advanced computer science — DSA, networking, security and full-stack development — to college-level students. Founder of LucazSoft. Independently designs, builds and ships complete software systems: from pharmacy supply-chain platforms to agentic AI runtimes. Currently competing in two active hackathons building on Solana and AMD Dev Cloud. Certified Ethical Hacker (CEH). MBA in progress.

TECHNICAL SKILLS

- AI & Agents:** LangChain, TinyFish Agent API, LLM Integration (Claude/Gemini), Agentic Workflow Design
- Blockchain:** Solana (Anchor/Rust), SPL Token, Token-2022, PDA Vaults, zkTLS (Reclaim Protocol), IPFS
- Frontend:** React.js, Next.js 15, TypeScript, Vite, Tailwind CSS, React Native, Flutter
- Backend:** Node.js, Express, FastAPI (Python), Go, GraphQL, REST APIs, WebSockets
- Databases:** MongoDB, Supabase/PostgreSQL, Firebase, SQLite, AES-256-GCM Encryption
- DevOps:** Docker, Vercel, Railway, AWS, Git/GitHub, CI/CD
- Security:** CEH (EC-Council), OAuth 2.0, Vulnerability Assessment, Network Infrastructure
- Teaching:** DSA, Advanced Networking, Cybersecurity, Full-Stack Dev, 5 years, 4 institutions

ACTIVE HACKATHON PROJECTS (May 2026)

Hackathon	Project	Role	Status
Solana Colosseum Frontier 2026	AETHER-LOGOS	Solo — Full-Stack + Rust	In Progress
AMD Developer Hackathon 2026	FORGE	Solo — Full-Stack + AI Agent	In Progress

AETHER-LOGOS — Asset-Light Trade Settlement Protocol · Solana Colosseum Frontier 2026

Eliminates trade finance friction using zkTLS proofs and atomic USDC escrow on Solana.

- Architected a two-program Anchor/Rust system (1,200+ lines): a 7-state trade escrow with USDC PDA vaults and a parimutuel prediction market for logistics risk hedging.
- Integrated zkTLS via Reclaim Protocol — converts live carrier data (DHL, FedEx, UPS, Maersk) into on-chain cryptographic Proofs of Custody that trigger permissionless USDC release.
- Built a Go microservice (~900 lines) that polls carrier APIs every 30 seconds, generates zkTLS proofs, and submits them to Solana via JSON-RPC.
- Next.js 15 / React 19 frontend with Phantom & Solflare wallet adapters, buyer/seller dashboards, real-time trade timelines, and IPFS invoice storage.

Stack: Rust · Anchor 0.32 · Solana · Go 1.22 · Next.js 15 · zkTLS · IPFS/Pinata · DHL API

FORGE — Autonomous Compute Exchange · AMD Developer Hackathon 2026

Agentic spot-price optimizer that reduces GPU spend on AMD Dev Cloud through intelligent batching and X402 micropayments.

- Three-agent runtime (Watcher / Strategist / Executor) powered by LangChain; agents cooperate on live AMD spot-price signals to make batch, wait or run-now execution decisions.
- FastAPI backend with WebSocket hub streaming live agent decisions and settlement receipts to dashboard in real time.
- Vite + React 18 + TypeScript dashboard with Recharts panels; X402 micropayment protocol for per-job transparent settlement.

Stack: *FastAPI · Python 3.11 · LangChain · AMD Dev Cloud · X402 · WebSockets · Vite · React 18 · TypeScript · Docker · Railway*

INDEPENDENT PROJECTS

DrPharmas — *Pharmacy Supply Chain Ecosystem* — *LucazSoft*

- Multi-tenant platform connecting manufacturers, dealers and retail pharmacies; real-time inventory sync across distributed nodes.
- Supabase/PostgreSQL for relational integrity; MongoDB for flexible product metadata; AES-256 encrypted authentication for sensitive medical data. Dockerised with tests.

Stack: *React.js · Node.js · Supabase · PostgreSQL · MongoDB · AES-256 · Docker*

Ghargharmaa — *Hyper-Local Marketplace & Logistics* — *Biratnagar*

- Full-stack local delivery and vendor management platform with dynamic pricing, diverse vendor catalogs, and real-time order tracking.
- Built solo after a co-founder exit; demonstrates end-to-end product ownership from architecture through deployment.

Stack: *Next.js · Node.js · MongoDB · TypeScript · Tailwind CSS*

GadiSewa — *Automotive Service & Supply Chain Platform*

- Cross-platform mobile app for garage management and parts ordering; real-time tracking optimised for low-bandwidth environments in Nepal.

Stack: *React Native · Node.js · MongoDB · REST APIs · Docker*

EXPERIENCE

Founder & Independent Developer | [LucazSoft \(Registered\)](#) 2023 – Present

- Sole designer, developer and maintainer of DrPharmas, Ghargharmaa and GadiSewa — three production-grade systems built end-to-end without a team.
- Currently competing solo in two international hackathons simultaneously (Solana Colosseum Frontier and AMD Dev Cloud, May 2026).

Technical Trainer & Faculty | [ISM T College](#) 2021 – Present

- Full-time faculty for 3.5 years; teaches Full-Stack Development, NoSQL database design and AI-augmented automation workflows to college students.
- Previously taught concurrently at 4 institutions: Himalayan Darshan (networking, security, web dev), ISMT, Adarsha School (DSA), and a +2 level college.
- Delivered 10+ workshops on agentic system design and AI-augmented development to external colleges and tech communities.

IT Officer | [Everest School](#) 2020 – 2021

- Managed school network infrastructure, automated administrative workflows, and ran online examinations during COVID-19 remote learning period.

Technical Recruiter (Part-time) | [Independent](#) 2019 – 2020

- Worked night-shift recruiting role in parallel with teaching to supplement income during early career period.

EDUCATION & CERTIFICATIONS

MBA — Strategic Management | [SASTHM](#)

In Progress

B.Tech — Computer Science & Engineering | [JNTUK, India](#)

2014 – 2018

Certified Ethical Hacker (CEH) · EC-Council / CTEVT (completed during +2 level)

WHAT I BRING

- Solo end-to-end ownership — I have shipped complete systems alone, from database schema to deployment, under real resource constraints.
- Agentic AI at the frontier — actively building multi-agent runtimes (LangChain, TinyFish) and blockchain-integrated AI systems in live competition environments.
- 5 years teaching complex CS topics means I can communicate clearly, document well, and onboard others — valuable beyond just writing code.
- Resilience — built across difficult personal and financial circumstances in Nepal, without institutional support or a team safety net.